Carrier Accumulation in GHz-Burst Femtosecond Excitation of Silicon

Burak Arayici March 23, 2026

1 The Problem: Unphysical Carrier Accumulation

In the burst-mode excitation scheme, N femtosecond pulses are delivered within a burst window τ_{burst}. Each pulse generates free carriers via two-photon absorption (TPA). The free-carrier density evolves according to Eq. 5 of the manuscript:

$$\frac{dN_{\text{fc}}}{dt} = \frac{\beta I^2}{2\hbar\omega} - \frac{N_{\text{fc}}}{\tau_c} \tag{1}$$

where β is the TPA coefficient, I is the pulse intensity, $\hbar\omega$ is the photon energy, and τ_c is the carrier lifetime.

Each femtosecond pulse generates a carrier density increment:

$$\Delta N_{\text{fc}} = \frac{\beta\, I_p^2\, \tau_p}{2\hbar\omega} \tag{2}$$

With our parameters ($\beta = 1.5 \times 10^{-11}\,\text{m/W}$, $I_p = 1 \times 10^{16}\,\text{W/m}^2$ i.e. $1\,\text{TW/cm}^2$, $\tau_p = 100\,\text{fs}$, $\hbar\omega = 1.87 \times 10^{-19}\,\text{J}$):

$$\Delta N_{\text{fc}} = 4.0 \times 10^{26}\,\text{m}^{-3} \quad \text{per pulse}$$

1.1 Figure 1: No recombination

If we neglect recombination entirely ($\tau_c \to \infty$), the carrier density after k pulses is simply (Eq. 6 of the manuscript):

$$N_k = k \cdot \Delta N_{\text{fc}} \tag{3}$$

This is a volumetric carrier density inside the silicon material itself — no assumption about particle geometry or concentration is needed. Figure 1 plots N_k vs. pulse number k for $N = 1000$ pulses.

The key comparison is against the **silicon atomic density**:

$$n_{\text{Si}} = \frac{8}{a^3} \approx 5.0 \times 10^{28}\,\text{m}^{-3}$$

where $a = 5.431\,\text{Å}$ is the lattice constant of the diamond cubic structure (8 atoms per unit cell). The carrier density cannot physically exceed the number of atoms available to donate electrons.

At the focus. All calculations use the Beer–Lambert attenuated intensity at $z = 500\,\mu\text{m}$ (see Section 1.2 for the BL derivation):

$$\Delta N_{\text{fc}}^{(\text{focus})} = \frac{\beta\, I_{\text{focus}}^2\, \tau_p}{2\hbar\omega} = 1.4 \times 10^{24}\,\text{m}^{-3} \quad \text{per pulse}$$

After $k = 1000$ pulses with no recombination: $N_{1000} = 1.4 \times 10^{27}\,\text{m}^{-3}$ ($0.03 \times n_{\text{Si}}$). At the focus the no-recombination model remains below n_{Si}, but gives a carrier density that grows without bound — the question is how much recombination reduces it further.

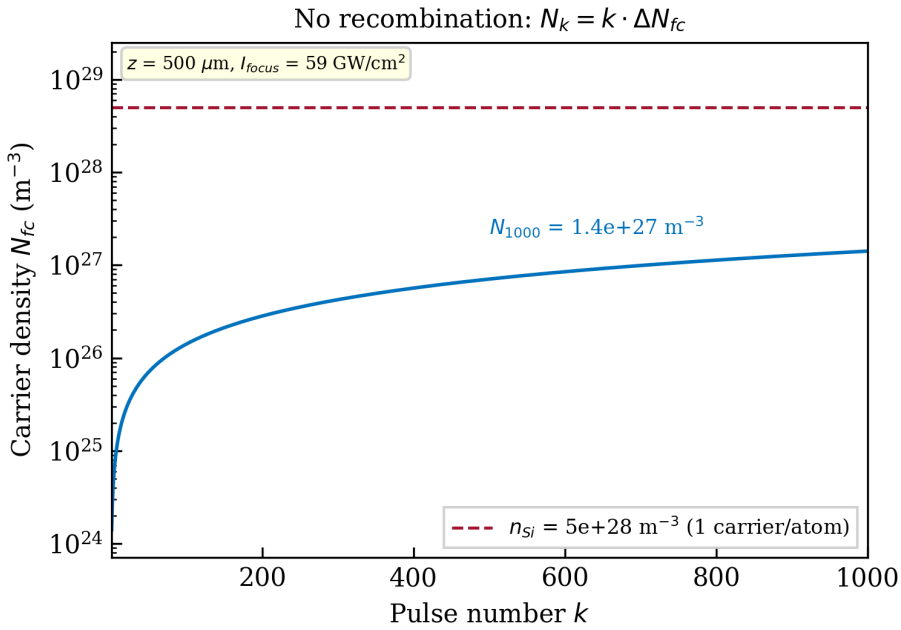

Figure 1: Carrier density vs. pulse number with no recombination at the focus ($z = 500\,\mu$m, $I_{\text{focus}} = 59\,\text{GW/cm}^2$). Without any recombination mechanism, carriers accumulate linearly to $N_{1000} = 1.4 \times 10^{27}\,\text{m}^{-3}$.

1.2 Figure 2: Full solution of Eq. 5 with pulsed excitation

The source term in Eq. (1) is not constant — it is a periodic step function. The laser is on (at intensity I) for duration τ_p, then off for the inter-pulse gap ($\tau_r - \tau_p$):

$$G(t) = \begin{cases} G = \dfrac{\beta I^2}{2\hbar\omega} & \text{during pulse} \\ 0 & \text{between pulses} \end{cases}$$

During the pulse ($0 < t < \tau_p$): The ODE is $dN/dt + N/\tau_c = G$, a first-order linear ODE. Using the integrating factor $\mu = e^{t/\tau_c}$:

$$\frac{d}{dt}\left[N\, e^{t/\tau_c}\right] = G\, e^{t/\tau_c}$$

Integrating with initial condition $N(0) = N_0$ (carrier density at the start of the pulse):

$$N(t) = G\,\tau_c\left(1 - e^{-t/\tau_c}\right) + N_0\, e^{-t/\tau_c} \tag{4}$$

At the end of the pulse ($t = \tau_p$):

$$N_+ = G\,\tau_c\left(1 - e^{-\tau_p/\tau_c}\right) + N_0\, e^{-\tau_p/\tau_c} \tag{5}$$

Between pulses ($\tau_p < t < \tau_r$): No generation, pure decay:

$$N(t) = N_+\, e^{-(t-\tau_p)/\tau_c} \tag{6}$$

At $t = \tau_r$ (just before the next pulse):

$$N(\tau_r) = N_+\, e^{-(\tau_r-\tau_p)/\tau_c}$$

Steady-state condition. In steady state, the carrier density must be periodic: $N(\tau_r) = N_0$. Substituting Eq. (5):

$$N_0 = \left[G\,\tau_c\Big(1 - e^{-\tau_p/\tau_c}\Big) + N_0\,e^{-\tau_p/\tau_c}\right] e^{-(\tau_r-\tau_p)/\tau_c}$$

Noting that $e^{-\tau_p/\tau_c} \cdot e^{-(\tau_r-\tau_p)/\tau_c} = e^{-\tau_r/\tau_c}$:

$$N_0\Big(1 - e^{-\tau_r/\tau_c}\Big) = G\,\tau_c\Big(1 - e^{-\tau_p/\tau_c}\Big)\,e^{-(\tau_r-\tau_p)/\tau_c}$$

$$\boxed{N_0 = \frac{G\,\tau_c\big(1 - e^{-\tau_p/\tau_c}\big)\,e^{-(\tau_r-\tau_p)/\tau_c}}{1 - e^{-\tau_r/\tau_c}}} \tag{7}$$

The peak density (right after a pulse) is:

$$N_{\text{peak}} = N_0\,e^{-\tau_p/\tau_c} + G\,\tau_c\Big(1 - e^{-\tau_p/\tau_c}\Big) \tag{8}$$

Simplification. Since $\tau_p = 100\,\text{fs} \ll \tau_c = 300\,\text{ps}$, we have $e^{-\tau_p/\tau_c} \approx 1$ and $1 - e^{-\tau_p/\tau_c} \approx \tau_p/\tau_c$, so:

$$N_{\text{peak}} \approx \frac{G\,\tau_p}{1 - e^{-\tau_r/\tau_c}} = \frac{\Delta N_{\text{fc}}}{1 - e^{-\tau_r/\tau_c}}$$

which recovers the discrete pulse-by-pulse result.

Intensity at the focus. The focal intensity I_p is set by the pulse energy, pulse duration, and focusing optics. With $F_p = 0.1\,\text{J/cm}^2$ and $\tau_p = 100\,\text{fs}$:

$$I_p = \frac{F_p}{\tau_p} = 1 \times 10^{16}\,\text{W/m}^2 \quad (1\,\text{TW/cm}^2)$$

Beer–Lambert attenuation. In tissue, the intensity at focal depth z is:

$$I(z) = I_p\,e^{-\mu_t\,z}$$

where $\mu_t = \mu_a + \mu_s = 5650\,\text{m}^{-1}$ [5]. At $z = 500\,\mu\text{m}$:

$$I_{\text{focus}} = 0.059\,I_p = 5.9 \times 10^{14}\,\text{W/m}^2 \quad (59\,\text{GW/cm}^2)$$

The per-pulse carrier generation at the focus:

$$\Delta N_{\text{fc}}^{(\text{focus})} = \frac{\beta\,I_{\text{focus}}^2\,\tau_p}{2\hbar\omega} = 1.4 \times 10^{24}\,\text{m}^{-3}$$

The steady-state peak density:

$$N_{\text{peak}} = \frac{1.4 \times 10^{24}}{1 - e^{-1/300}} = 4.2 \times 10^{26}\,\text{m}^{-3} \quad (0.008 \times n_{\text{Si}})$$

This is well below n_{Si} — the combination of Beer–Lambert attenuation and the correct pulsed duty cycle ($\tau_p/\tau_r = 0.1$) keeps the carrier density physical at this depth.

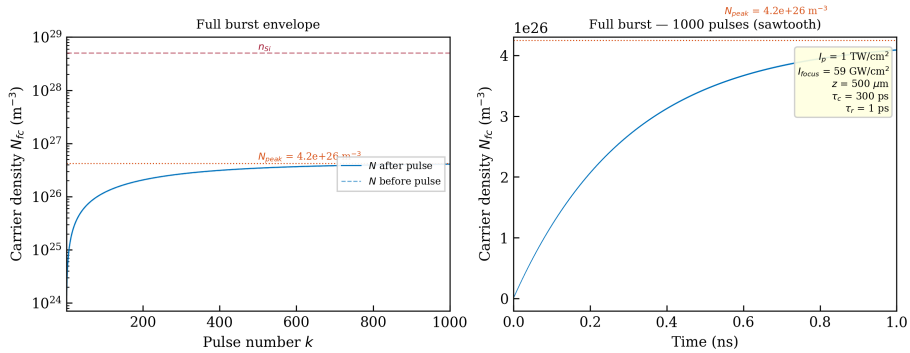

Figure 2: Pulsed ODE solution at the focus ($z = 500\,\mu$m, $I_{\text{focus}} = 59\,\text{GW/cm}^2$). The carrier density shows the sawtooth build-up and inter-pulse decay, reaching a steady-state peak of $N_{\text{peak}} = 4.2 \times 10^{26}\,\text{m}^{-3}$.

2 Adding Auger Recombination

We simply add one term to Eq. (1):

$$\frac{dN}{dt} = \frac{\beta I^2}{2\hbar\omega} - \frac{N}{\tau_c} - C\,N^3 \tag{9}$$

where $C = 3.8 \times 10^{-31}\,\text{cm}^6/\text{s}$ is the Auger coefficient for c-Si [4]. The only change is the CN^3 term.

Between pulses ($G = 0$): the ODE becomes:

$$\frac{dN}{dt} = -\frac{N}{\tau_c} - C\,N^3 \tag{10}$$

This is a Bernoulli equation. We solve it by substituting $u = N^{-2}$, which gives $du/dt = -2\,N^{-3}\,dN/dt$:

$$\frac{du}{dt} = \frac{2u}{\tau_c} + 2C$$

This is linear in u and solvable with the integrating factor e^{-2t/τ_c}:

$$u(t) = (u_0 + C\,\tau_c)\,e^{2t/\tau_c} - C\,\tau_c$$

Substituting back $u = 1/N^2$ with initial condition $N(0) = N_0$:

$$\boxed{N(t) = \frac{1}{\sqrt{\left(\frac{1}{N_0^2} + C\,\tau_c\right)e^{2t/\tau_c} - C\,\tau_c}}} \tag{11}$$

Same stepwise framework. Since $\tau_p \ll \tau_c$, decay during the pulse is still negligible. The algorithm is identical to before:

1. Pulse arrives: $N \to N + \Delta N_{\text{fc}}$
2. Between pulses: apply Eq. (11) with $t = \tau_r$

The only change is replacing $\exp(-\tau_r/\tau_c)$ with Eq. (11).

Result at focus ($z = 500\,\mu$m). Figure 4 compares all three models at the same focal intensity ($I_{\text{focus}} = 59\,\text{GW/cm}^2$):

- **No recombination**: $N_{1000} = 1.4 \times 10^{27}\,\text{m}^{-3}$ — unbounded linear growth.
- **SRH (surface) only**: $N_{\text{ss}} = 4.2{\times}10^{26}\,\text{m}^{-3}$ — a 3.3× reduction from the no-recombination case.
- **SRH + Auger**: $N_{\text{ss}} = 1.4 \times 10^{26}\,\text{m}^{-3}$ — an additional 3× reduction from the Auger channel, or 10× total reduction from the no-recombination case.

All three models remain below n_{Si} at the focal depth, but the progressive reduction demonstrates the importance of including both recombination mechanisms.

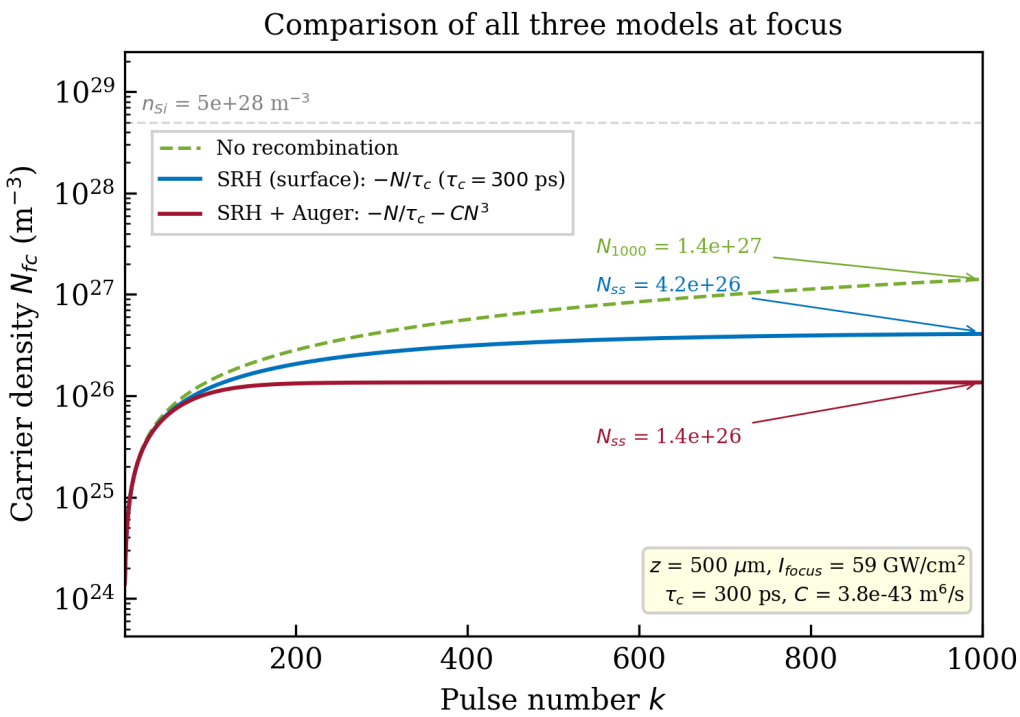

Figure 3: Comparison of all three models at the focus ($z = 500\,\mu$m, $I_{\text{focus}} = 59\,\text{GW/cm}^2$): no recombination (dashed), SRH surface recombination only ($\tau_c = 300\,\text{ps}$), and SRH + Auger ($C = 3.8 \times 10^{-31}\,\text{cm}^6/\text{s}$). Each mechanism progressively reduces the steady-state carrier density.

3 Summary

Three arguments establish the carrier dynamics:

1. **Without recombination**, carriers accumulate linearly (Eq. 6). At the focus, $N_{1000} = 1.4 \times 10^{27}\,\mathrm{m}^{-3}$ $(0.03 \times n_{\mathrm{Si}})$.

2. **With SRH surface recombination** (Eq. 5, $\tau_c = 300\,\mathrm{ps}$), the pulsed ODE gives a finite steady state: $N_{\mathrm{ss}} = 4.2 \times 10^{26}\,\mathrm{m}^{-3}$ $(0.008 \times n_{\mathrm{Si}})$ — a 3.3× reduction. The linear N/τ_c term represents SRH recombination at surface trap states [3], with $\tau_c = d/4S$, experimentally confirmed via $\tau \propto d$ in Si nanostructures [1, 2]. At their carrier densities ($\sim 10^{19}\,\mathrm{cm}^{-3}$), the Auger lifetime $\tau_A = 1/CN^2 \approx 26\,\mathrm{ns}$ is ∼100× longer than the measured surface lifetime, confirming Auger was negligible in those experiments.

3. **With SRH + Auger** $(N/\tau_c + CN^3)$, the steady state drops to $N_{\mathrm{ss}} = 1.4 \times 10^{26}\,\mathrm{m}^{-3}$ $(0.003 \times n_{\mathrm{Si}})$ — an additional 3× reduction, or 10× total from the no-recombination case. At our per-pulse carrier densities ($\sim 10^{20}\,\mathrm{cm}^{-3}$), the Auger lifetime $\tau_A \approx 16\,\mathrm{ps}$ is comparable to the inter-pulse spacing, making this channel significant.

Model	N_{ss} (m^{-3})	$N_{\mathrm{ss}}/n_{\mathrm{Si}}$	Reduction	Equation
No recombination ($k = 1000$)	1.4×10^{27}	0.03	1× (ref)	Eq. (3)
SRH surface ($\tau_c = 300\,\mathrm{ps}$)	4.2×10^{26}	0.008	3.3×	Eq. (7)
SRH + Auger ($\tau_c + CN^3$)	1.4×10^{26}	0.003	10×	Eq. (11)

Table 1: Steady-state carrier density at the focus ($z = 500\,\mu\mathrm{m}$, $I_{\mathrm{focus}} = 59\,\mathrm{GW/cm}^2$) for each model. The "Reduction" column shows the factor relative to the no-recombination case. Parameters: $\tau_c = 300\,\mathrm{ps}$, $C = 3.8 \times 10^{-31}\,\mathrm{cm}^6/\mathrm{s}$, $\mu_t = 5650\,\mathrm{m}^{-1}$.

References

[1] E. M. Grumstrup *et al.*, "Ultrafast carrier dynamics of silicon nanowire ensembles," J. Phys. Chem. C **118**, 8634 (2014).

[2] H. Li *et al.*, "Nanoscale imaging of carrier dynamics in individual silicon nanowires," Nano Lett. **23**, 1445 (2023).

[3] S. S. Li, *Semiconductor Physical Electronics*, 2nd ed. (Springer, 2006), Ch. 6.

[4] J. Dziewior and W. Schmid, "Auger coefficients for highly doped and highly excited silicon," Appl. Phys. Lett. **31**, 346 (1977).

[5] A. N. Yaroslavsky *et al.*, "Optical properties of selected native and coagulated human brain tissues," Phys. Med. Biol. **47**, 2059 (2002).